

Comparative Aerodynamic Analysis Of Wing Planforms

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Abstract

This is a comparative RANS study of planform effects on wing aerodynamics at subsonic speeds (30ms^{-1}) between three different planforms: Rectangular, Tapered & Swept-Back Constant Chord. The wings were designed consistently following the NACA 2412 airfoil profile on both the root chords & tip chords. OnShape was used for CAD modeling, OpenFOAM for the CFD simulation, MATLAB & ParaView for post-processing. Short python scripts were also used in order to support this project.

CFD configuration: Incompressible, steady-state, $k - \omega$ SST turbulence model (Reynolds-averaged Navier-Stokes equations), 5 boundary layers @ 1.20 growth rate, constant fluid properties.

Key Performance Indicators (KPIs): Lift Coefficient (C_L) & Drag Coefficient (C_D) relative to 5 deg freestream flow (simulating AoA)

Of the three, the tapered wing produced the highest C_L/C_D ratio at 6.14. The swept-back wing showed the worst performance profile of all three (theoretically expected due to subsonic speeds), although not very far from the rectangular wing.

Author's Note: This project was carried out independently as a pre-university student, self-taught in CAD and CFD, with no formal CFD coursework or lab supervision at the time of writing. Additionally, all meshing & solving was performed on a machine with 16GB RAM under WSL. These constraints directly drove multiple methodology decisions explained in later sections of this document. This study is meant to produce comparative results, not absolute accurate theoretical results for each individual coefficient.

1 Methodology

1.1 Geometry & CAD

NACA 2412 airfoil .dat coordinates were outsourced from the internet. A short python script was then developed to fetch the coordinates in the file and plot it using matplotlib. The airfoil coordinates are then scaled as per requirement (0.2 m for rectangular & swept configurations, 0.267 m & 0.133 m for the tapered configuration). A .dxf was then generated and imported to OnShape, connecting the plotted coordinates with splines, then lofted as required. The design drawings for all three of the planforms are shown below. The drawings were generated using OnShape's built-in features. For a quick reference, the tapered wing inhibits a taper ratio of 0.5, equally tapered from both the LE & TE, lofted using a mid-chord reference point. The swept-back wing inhibits a 30° sweep angle.

Some calculations might require magnification. This was done to ensure not too much space was taken over by the drawings.

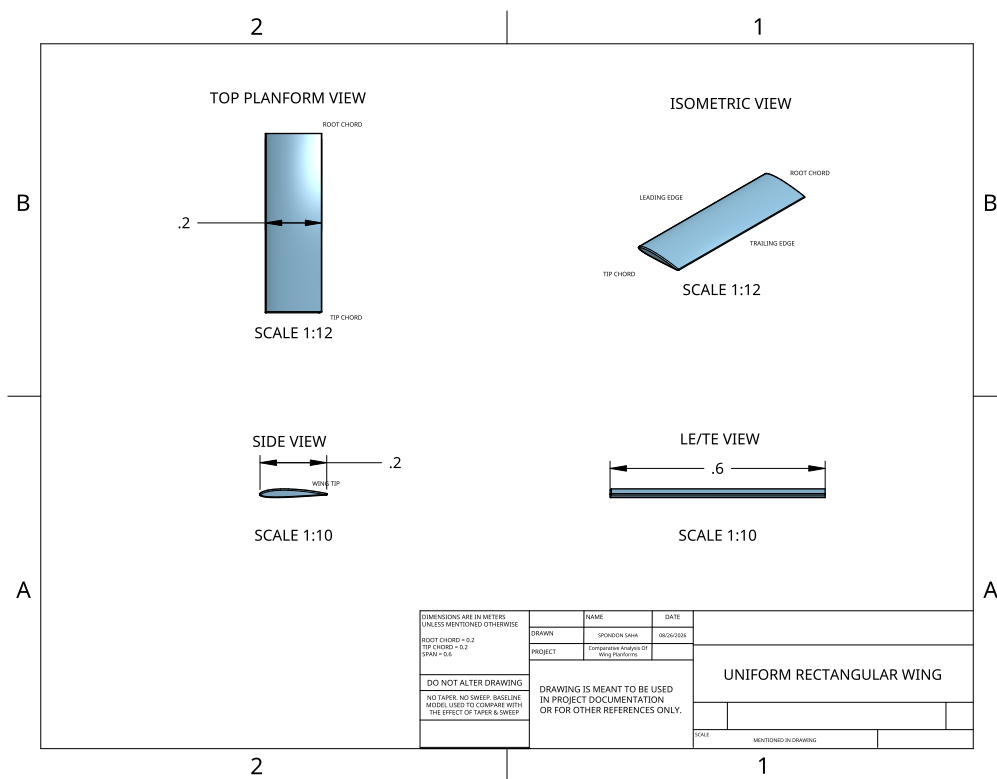


Figure 1: Rectangular Wing

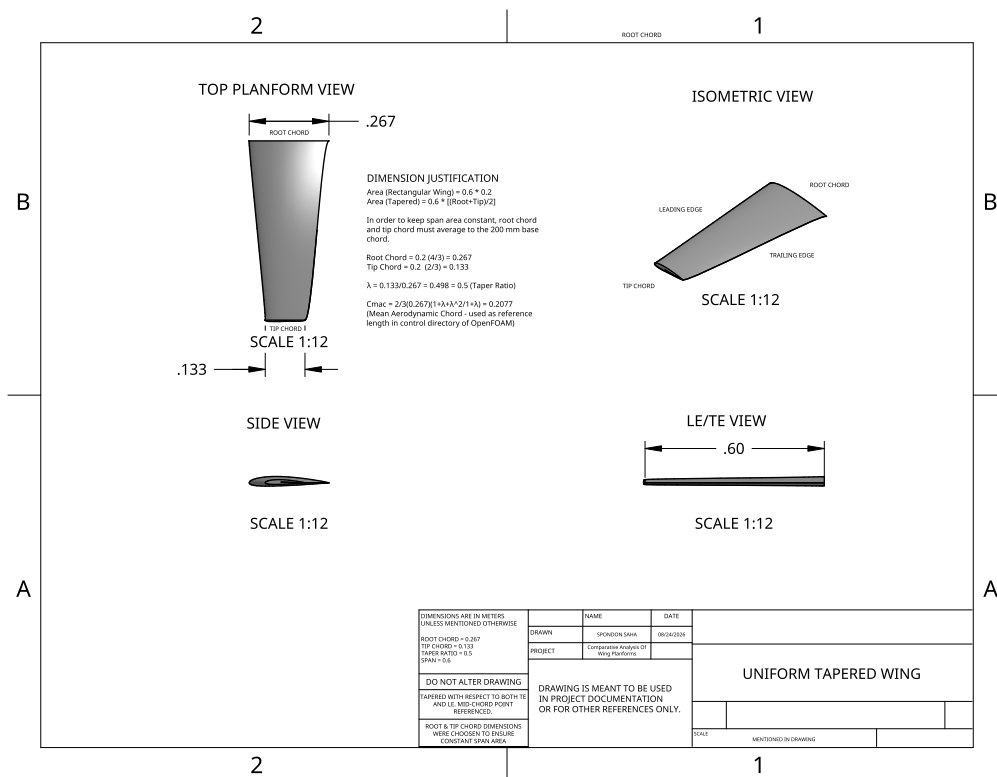


Figure 2: Tapered Wing

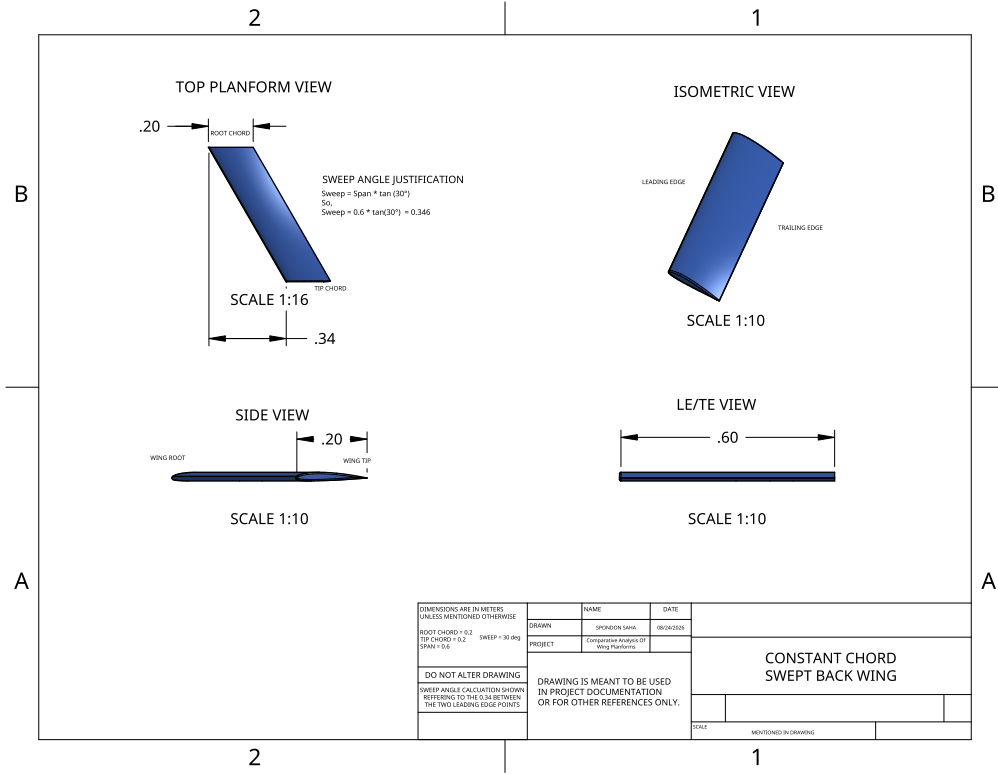


Figure 3: Swept-Back Constant Chord Wing

1.2 Computational Domain & Wall Functions

Standard domain dimensions were used for this project.

5 chord lengths upstream.

10 chord lengths downstream. *One small caveat — the sweep angle of 30 deg on the swept back wing effectively reduces the downstream length of the domain, but it is an insignificant difference for the purpose of this study.*

5 chords lengths top & bottom.

3 chord lengths from wingtip.

Root chord was attached to the wall of the domain.

All walls except the wall attached to the root of the wing was defined as slip walls. The mentioned wall was defined as "symmetryPlane". This was done majorly to simulate the presence of a fuselage. The Aref (reference area) was listed as $0.12m^2$ inside controlDict (OpenFOAM file) to correct the calculated lift & drag values for a wing with $AR = 3$. While this was done to simulate the effect of a fuselage, it does not capture the fairing flow or the fuselage induced upwash ahead of the wing. This is purely to avoid tip vortices forming at the root of the wing. Additionally, omegaWallFunction (turbulence specific dissipation rate), kqRWallFunction (turbulent kinetic energy) & nutUSpalding-WallFunction (turbulent kinematic viscosity, ν) were used. *The calculated values below & standard values above were used in the CFD configuration files where appropriate.*

Air Density = $1.225 kgm^{-3}$

Dynamic Viscosity = $1.7894 \times 10^{-5} m^2s^{-1}$

Freestream Velocity = $30 ms^{-1}$

$$\nu = \frac{\mu}{\rho} = \frac{1.7894 \times 10^{-5}}{1.225} = 1.4607 \times 10^{-5} m^2s^{-1}$$

$$k_{inlet} = \frac{3}{2}(30 \times 0.03)^2 = 1.215 m^2/s^2$$

3 % turbulence was simulated

$$\omega_{inlet} = \frac{\sqrt{1.215}}{(0.09)^{1/4}(0.07(0.2))} \approx 143.75 \text{ s}^{-1}$$

0.09 is the C_μ (Turbulent viscosity constant). $0.07 \times \text{Chord length}$ is calculating L_{turb} (turbulent length scale)

1.3 Meshing Strategy

The mesh was extracted from OnShape as an .stl, then placed inside /triSurface. The domain was created using blockMesh, with 50 mm base cells. snappyHexMesh (castellatedMesh + snap) was then used to snap the .eMesh into the domain. The .eMesh was created using surfaceFeatureExtract. A refinement level of 7 was selected as a sweet spot between accuracy & computational limitations. Local refinement boxes were used around the wing in order to refine the mesh around the wing and towards the trailing edge. Furthermore, 5 boundary layers were added at a growth rate of 1.20. A y_1 sizing python script was developed in order to determine the first cell height, which was then placed inside the addLayers section.

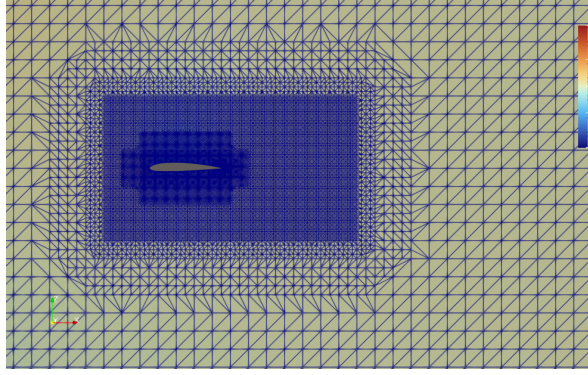


Figure 4: Domain visualization in ParaView

1.4 AoA Implementation Method

The freestream velocity for this project is 30 ms^{-1} , and the CAD model was not rotated during export. Accordingly, vectorization of the velocity was used in order to achieve the effect of 5 degree angle of attack on the wing under simulation. In mathematical vector form:

$$U_\infty = \begin{pmatrix} \cos 5^\circ U_\infty \\ 0 \\ \sin 5^\circ U_\infty \end{pmatrix}$$

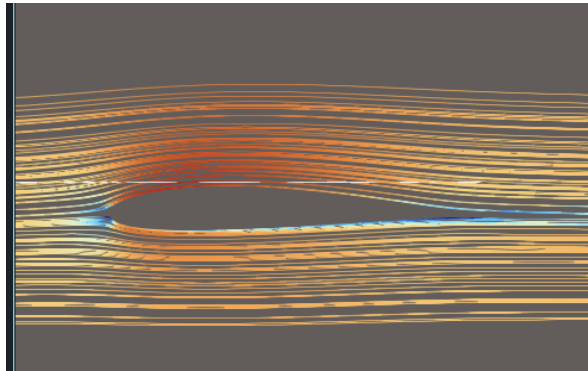


Figure 5: 5° flow visualization

2 Aerodynamic Summary

This section further analyzes the simulated aerodynamic coefficients of the three tested wing planforms to evaluate their performance under subsonic configurations.

2.1 Aerodynamic Performance Profiles

The table below shows a quick overview of the CFD simulation results on the models. These are the pillars of the comparative conclusion for this study.

Table 1: Aerodynamic Performance of Wing Geometries @ Subsonic Speeds

Wing Geometry	Lift Coefficient C_L	Drag Coefficient C_D	Ratio C_L/C_D
Rectangular	0.1602	0.0293	5.47
Tapered	0.2042	0.0332	6.15
Swept-Back Constant Chord	0.1516	0.0289	5.25

Coefficient values are rounded off to 4 decimal points, and ratio is rounded to 2 decimal points.

2.2 Baseline Configuration — Rectangular Wing

The rectangular wing represents the baseline configuration for this study, yielding moderate lift ($C_L = 0.1602$) & a moderate drag profile ($C_D = 0.0293$). The chord remains constant from root all the way to the tip, causing the wing to shed a significant amount of lift at the wingtips. As a direct consequence, the wingtip generates strong & concentrated tip vortices. This creates a large downwash component, directly resulting in a higher induced drag relative to the produced lift, restricting it's aerodynamic efficiency to a C_L/C_D value of 5.47

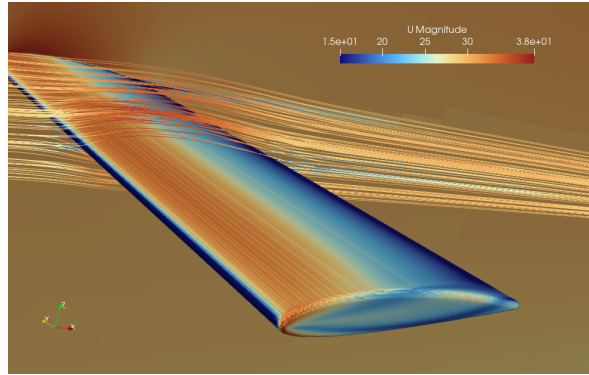


Figure 6: Streamlines over the rectangular showing velocity changes throughout the wing surface.

2.3 Best Aerodynamic Performance — Tapered Wing

The tapered wing configuration exhibited the highest lift at a C_L value of 0.2042 & the highest overall aerodynamic efficiency at a C_L/C_D value of 6.15. As this design reduced the chord length from the root to the tip, the spanwise lift distribution closely replicates that of an ideal elliptical lift distribution. This heavily optimizes the downwash velocity along the span, allowing the wing to generate more effective lift for a given surface area. Theoretically, the drag is supposed to be lower due to the more optimized spanwise loading, but the higher overall lift increases the strength of the trailing wingtip vortices, resulting in a slightly higher drag at $C_D = 0.0332$.

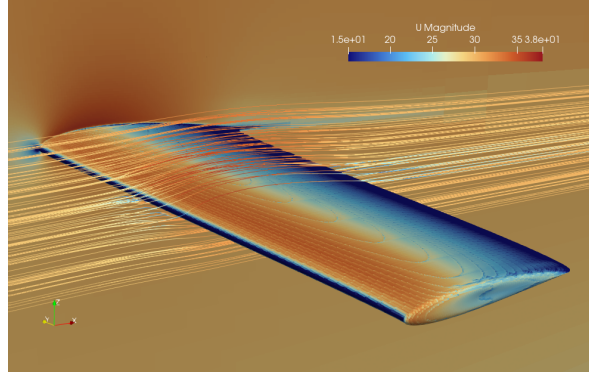


Figure 7: Streamlines over the tapered showing velocity changes throughout the wing surface.

2.4 Worst Aerodynamic Performance — Swept-Back Wings

The swept-back constant chord generated the lowest lift at a C_L value of 0.1516. This and its drag profile being only slightly lower than that of the rectangular wing resulted in the lowest subsonic efficiency between the three planforms. In a swept wing, the freestream velocity gets split into two vector components: Chordwise & Spanwise. Only the chordwise velocity contributes to the acceleration of the air over the airfoil upper surface, significantly reducing its effective lift generation. At subsonic speeds, wave drag is absent, leading to only a marginal reduction in C_D . At higher speeds, the drop in drag would be significant to give a significantly improved overall efficiency (C_L/C_D).

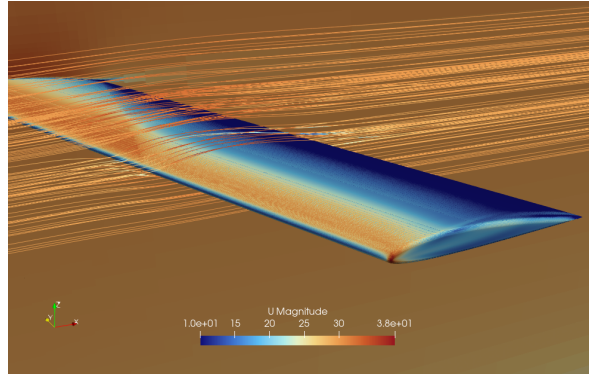


Figure 8: Streamlines over the swept-back showing velocity changes throughout the wing surface.

3 Validation Against Theoretical Aerodynamics

3.1 Pressure Differential Theory (Bernoulli's Principle)

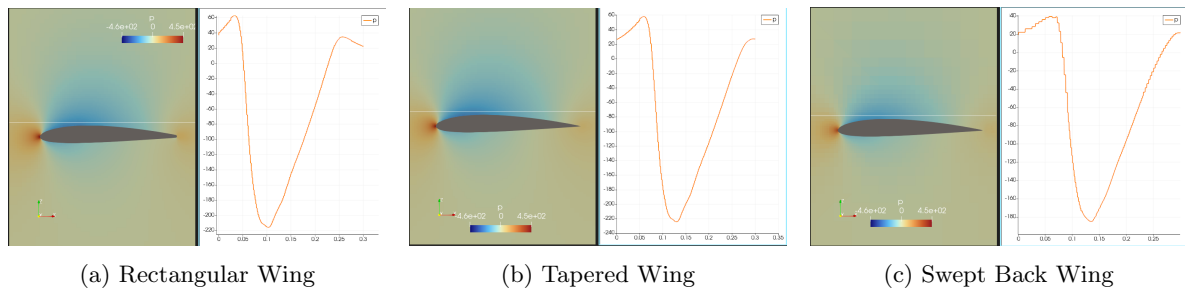


Figure 9: Chordwise Pressure Plot @ Mid Span

Bernoulli's principle states the low pressure suction created on the top of the wing and the high pressure area created at the bottom of the wing surface works together to create lift on the wing body. As shown is Figure 9, there is a increase in pressure at the leading edge (stagnation point, air is compressed), then the low pressure suction is present on all three of the wings immediately after. The lowest pressure of the tapered wing is noticeably lower than the other two, which directly correlates to the tapered wing showing a significantly higher lift profile— $\approx 27\%$ higher C_L than the rectangular configuration. Similarly, the lowest pressure of the swept-back wing is slightly higher than that of the rectangular wing, which again directly correlates to the lowest lift generation of this planform, ($\approx 5.7\%$) lower C_L than the rectangular planform.

The lower pressure is formed as a direct consequence of the acceleration of the freestream velocity that is caused by the shape of the airfoil. This was also visualized.

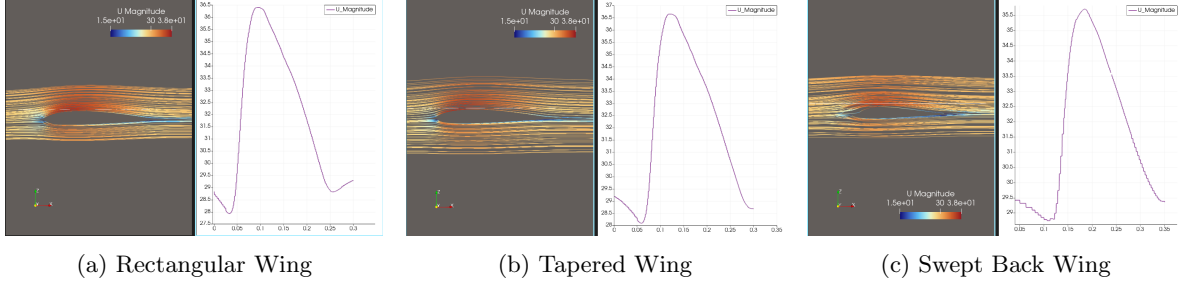


Figure 10: Chordwise Velocity Plot @ Mid Span

Similarly, the highest velocity of the tapered wing is significantly higher $\approx 36.7ms^{-1}$ than the other two planforms. At the suction area, streamlines converge, forcing a massive acceleration of the local fluid velocity. Consequently, the pressure drops below ambient levels, generating a net suction effect.

$$\Delta p = p_{lower} - p_{upper}$$

3.2 Generated Lift For Each Planform

$$Lift = (C_L)(S)(\rho/2)(v)^2$$

Calculated lift for a rectangular wing following the NACA 2412 airfoil & simulation results:

$$Lift = (0.1602)(0.12)(1.225/2)(30)^2 \approx 10.6N$$

Calculated lift for a tapered wing following the NACA 2412 airfoil & simulation results:

$$Lift = (0.2042)(0.12)(1.225/2)(30)^2 \approx 13.5N$$

Calculated lift for a swept-back constant chord wing following the NACA 2412 airfoil & simulation results:

$$Lift = (0.1516)(0.12)(1.225/2)(30)^2 \approx 10.0N$$

Lift forces are comparable to that of a small UAV wing based on the low surface area of the model.

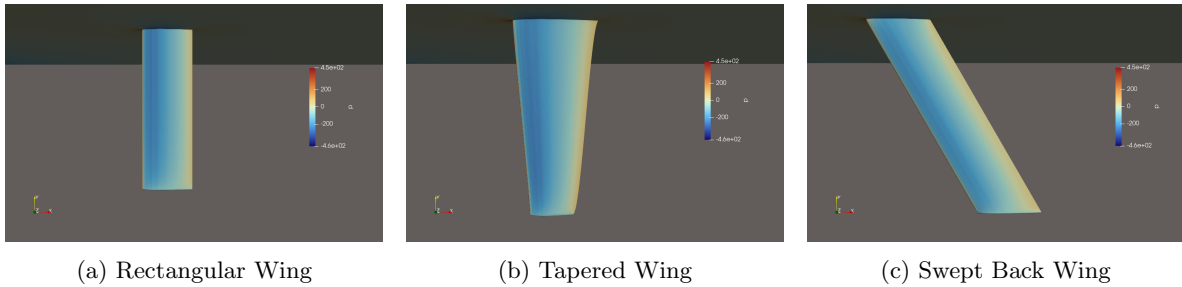


Figure 11: Pressure Contours— Top View

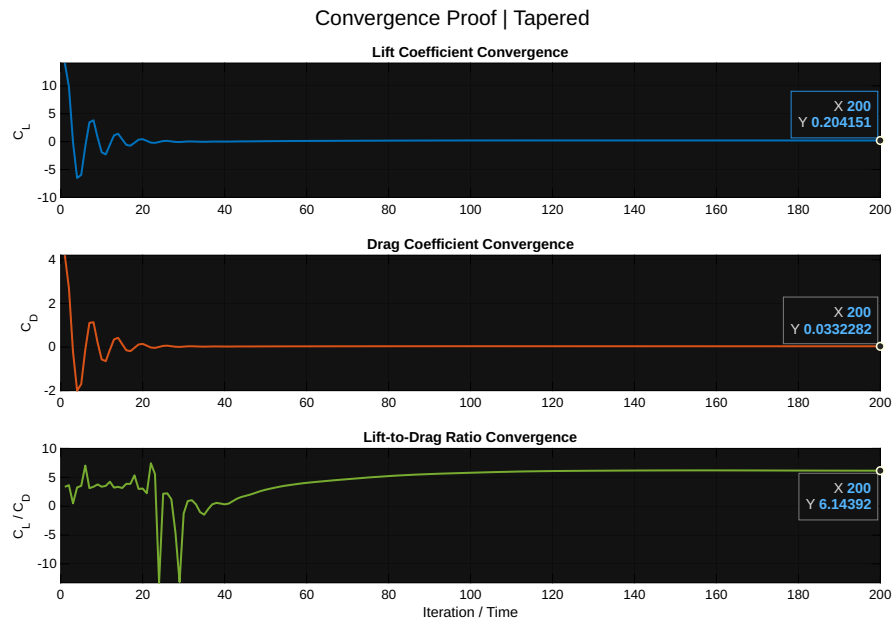
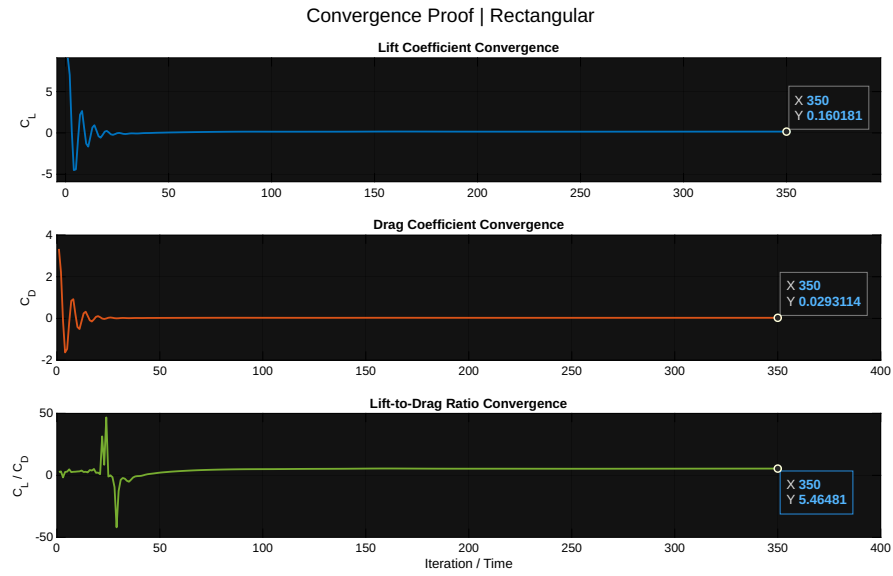
LE is to the left of the screen on each image

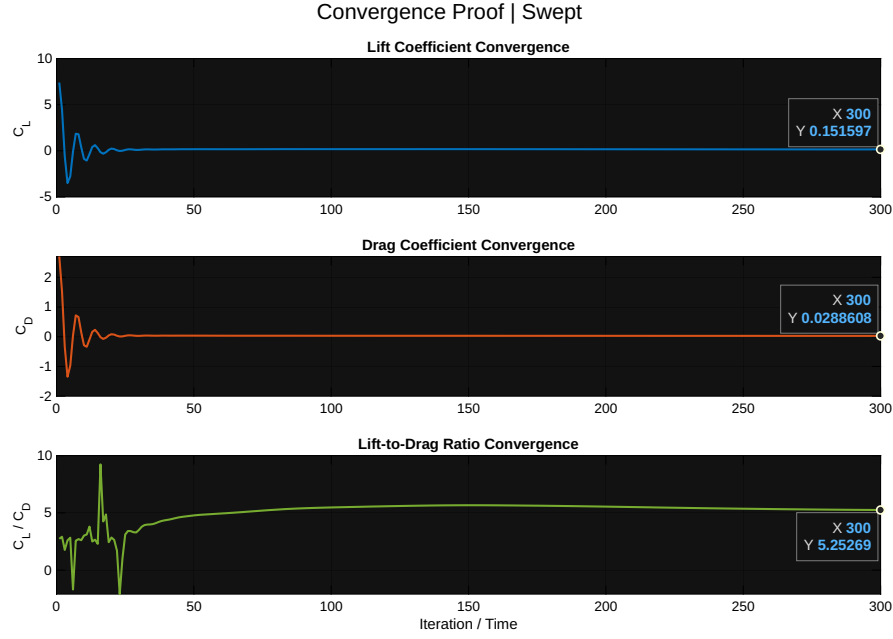
3.3 Newton's Third Law

When the air hits the leading edge of the wing and flows around the shape of the airfoil (Coandă Effect), the 2 % camber of the NACA 2412 airfoil causes the airflow to deflect downwards. This can be seen clearly in Figure 5. Newton's Third Law states that all actions have an equal & opposite reaction. As the airflow is pushed downwards, naturally, the wing is pushed upwards, which also generates lift. This also demonstrates why a higher AoA usually leads to a higher lift coefficient for as long as the Coandă effect is maintained/satisfied (Flow detachment would lead to a stall, which would significantly deteriorate lift generation).

4 Convergence Verification

For every CFD solver run, low residuals alone isn't enough to determine true convergence. Accordingly, a MATLAB script was developed in order to plot C_L , C_D , C_L/C_D over each iteration to show the values have plateaued out by latest iteration in each case.

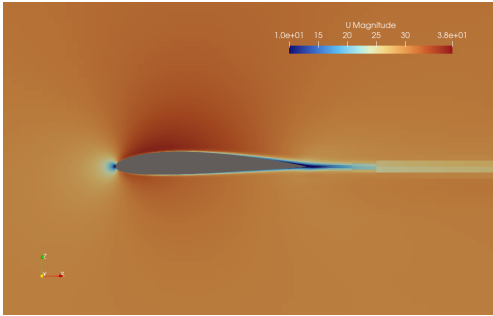




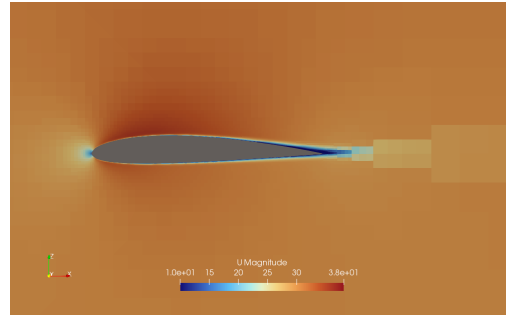
5 Study Conclusions & Considerations

5.1 Aerodynamic Conclusion

Which planform produces the best lift-to-drag ratio at a speed of 30ms^{-1} & an angle of attack at 5° ? Based on this study, tapered wing. This is theoretically defensible as the tapered wing narrows towards the tip, and so it approximates an elliptical wing's lift distribution, which is more optimized than that of a rectangular wing's distribution. Furthermore, the sweep angle did not yield much of a difference as it becomes significant only during transonic flights. At subsonic speeds, there are no advantages to the reduced drag that swept wings can generate. Instead, a spanwise velocity flow vector is now introduced, causing air to travel outward towards the wingtip.



(a) Tapered Wing Airflow Profile



(b) Swept-Back Wing Airflow Profile

Figure 12: Velocity Contours Showing Wake Area Flow

As per the ParaView visualizations above, this spanwise velocity vector inherently leads to earlier flow separation & a wider wake. Whereas, the tapered wing profile gives a smooth, streamlined wake as the Coandă effect is maintained further down the airfoil.

5.2 Project Files

All files related to this project— OpenFOAM file configurations, more visualizations, .stl files, a progress log documentation, & the code (MATLAB & Python) scripts that were used in the making of this project is available as a public repository in GitHub [here](#).

5.3 Error Sources & Limitations

- Low relative C_L values. This is suspected to be due to the low aspect ratio, but it is unlikely that alone would have caused a low C_L . Since this was a comparative study & the case structure was similar for all three, this caveat is negligible yet worth mentioning.
- No formal mesh independence verification study was performed. Avoided due to computational limitations, as each solver run took ≈ 1 day.
- Symmetry plane root does not accurately represent a fuselage as it lacks curvature & is classified as a slip wall in this study.
- This project is not heavily technical, it only explores introductory aerodynamic concepts & comparisons. Furthermore, it does not include the deeply technical mathematical derivations & justifications.
- As stated in the Author's Note, this study was conducted by a pre-university high school graduate with a particular interest in aerospace & aeronautics. It may contain errors & discrepancies that went completely unnoticed during the compilation of this project.

5.4 Future Scope

This project can be repeated with a transient flow simulation, & across multiple AoA values. The swept-wing configuration can be further studied with a transonic compressible flow condition to see its drag reducing effects in action in order to gain an increased depth of understanding about the concepts introduced in this study.

5.5 Software & Theory Stack

This study was conducted entirely using open-source free to use web & desktop applications. As the open-source CFD software OpenFOAM requires a deep technical understanding of the underlying physics, a list of important theoretical concepts is also included.

Software:

- OpenFOAM (Ubuntu WSL— Fluid Simulation)
- OnShape (Web— CAD Modeling)
- VS Code (IDE for Python Scripts & OpenFOAM configuration files)
- GitHub (Real Time Cloud Storage & Documentation)
- Git (Local Directory & Cloud Directory Collaboration)
- ParaView (CFD Post-Processing)
- MATLAB (Web— Data Analysis)
- Overleaf (LaTeX Documentation)

Theoretical Concepts:

- Aerodynamics Of Lift Generation
- Difference Between Rectangular, Tapered & Swept-Back Wing Configurations
- Introductory Fluid Dynamics
- Boundary Conditions, Turbulence Models & Wall Conditions
- Governing Navier-Stokes Equations & Their Derivations
- Pre-Processing, Solver & Post-Processing Stages Of A Fluid Simulation
- Visualization & Analysis Of CFD Results